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Project Title:	Biodegradable Polymers		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Biological physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Performance Goal No.:	1.24		
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The desire to create more functional nanoscale devices has driven scientists to redirect their attention to the development of bottom-up strategies that employ biological molecules. Bionanofabrication describes a novel fabrication process that takes advantage of the specificity and/or catalytic efficiency of biological systems to create novel nanoscale structures. As a result, a novel method of fabricated biopolymeric micro-/nano-structures will be developed and the underlying physical basis for their unique structure-function relationship will be elucidated. As a model system we will explore polyhydroxyalkanoates (PHAs) for which significant knowledge on the genetics and biochemistry exists. PHAs are a family of aliphatic polyesters produced by a variety of microorganisms as a reserve of carbon and energy. The enzymes involved in the synthesis of PHAs can be utilized to produce novel polymers in vitro, both, in bulk and on solid surfaces. Here, site-specific attachment of the key catalytic enzyme, PHA synthase, on lithographic fabricated surfaces and subsequent addition of 3-(R)-hydroxybutyryl-CoA substrates ((R)-HB-CoA), allowed us to create spatially ordered

Task Description:	polyhydroxybutyrate (PHB) polymeric nanostructures on micro-patterned surfaces via in situ enzymatic surface-initiated polymerization (ESIP). We believe that the use of this novel bionanofabrication approach offers many practical applications in different areas especially in biomedical applications. For example, the modification of solid surfaces with this biocompatible and biodegradable polymer could potentially yield novel coated implants for tissue engineering. 1.) Development of a novel surface polymerization strategies based on biodegradable and biocompatible PHAs under physiological conditions. 2.) Study the effect of various additives to enhance this in situ bionanofabrication method of polyhydroxybutyrate (PHB) micro-/nano-structures to create novel hybrid polymeric structures on solid surfaces 3.) To gain a better understanding of the surface growth mechanisms and the effect of additives on this particular ESIP process could facilitate its use in bionanofabrication applications. 4.) Use this understanding to rationally design and synthesize other novel polymeric structures 5.) Test their applications of the fabricated polymeric structures on various solid surfaces Solid surfaces modified with polymeric materials have been shown to have a wide range of applications, such as providing protective coatings, controlled adhesion, and improved biocompatibility. Among the methods for modifying solid surfaces, surface-initiated polymerization (SIP) has been intensively studied and shown to produce more robust and denser films than conventional spin coating or grafting. In SIP, the reactive group that initiates the polymerization is immobilized onto the surface, and the propagating polymer chain is grown directly from the surface. SIP is particularly attractive because it is compatible with micro- or nanolithography processes. Living polymerization to form polymer films on solid surfaces using radical, cationic, anionic and metal alkoxide-based initiation processes have been demonstrated for various monomers. However, more environmentally friendly and less hazardous catalysts are currently desired to broaden the applications of such biopolymers, especially in biomedical areas, where the minimization of harmful species is critical. In addtion, biopolymers are 'green' and can have a significant role as materials for long-term space flight reducing the environmental impact and potentially being a major structural component in a closed system. Thus, we have successfully developed a novel SIP system based on biological catalysts, enabling the fabrication of a biodegradable and biocompatible polymer, polyhydroxyalkanoate (PHA), on various solid surfaces such as agarose, silicon, and gold.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our novel strategy of surface polymerization of biopolymer PHAs can serve as environmentally friendly matrices for development of novel surface coating strategies.
Task Progress:	We have studied and developed an enhanced in situ bionanofabrication method of polyhydroxybutyrate (PHB) micro-/nano-structures to create novel hybrid polymeric structures on solid surfaces under physiological conditions. The method was conducted successfully via two steps: The first step involves an immobilization of PHA synthases to a solid surface using Ni-NTA chelating groups and the second step is a catalytic polymerization of monomeric substrate, (R)-3HB-CoA, by the immobilized synthases. In this study, the effect of BSA on stabilizing the immobilized enzyme and the PHB polymeric structure was found to be the most important factor for a successful surface polymerization in order to form thick PHB films (up to 1 mm in height) with full surface coverage. The resulting PHB surface structure obtained via BSA-stabilizing process is, to the best of our knowledge, the highest surface polymeric structure synthesized to date compared to other SIP techniques using biological catalysts. We believe that the use of this novel bionanofabrication approach offers many practical applications in different areas. For example, the modification of solid surfaces with this biocompatible and biodegradable polymer could potentially yield novel coated implants for tissue engineering that promote cell attachment and growth. As a result, one of our future goals is to study how different cells interact on this type of fabricated PHB surface. In addition, we plan to employ ESIP for in situ synthesis of other PHA micro-/nano-structures by using different PHA synthases or substrates to generate polymers with desired characteristics and, potentially, novel compositions and structures (e.g. random, block copolymers or the incorporation of specific functional groups). Besides biocompatible surface coatings, we envision that the novel polymeric micro-/nano-structures can be built in spaces that cannot be accessed by convention lithographic tools or other fabrication process. For example, PHB microstructures could be formed in situ inside microfluidic channels on designated areas and potentially used as either microfluidic valves or pumps or passive micromixers inside microfluidic channels. Finally, we believe that the information obtained through this study also helps extending our understanding of kinetics and mechanisms of in vitro solid-phase PHA synthesis.
Bibliography Type:	Description: (Last Updated: 09/03/2009)
Abstracts for Journals and Proceedings	Niamsiri N, Bergkvist M, Delamarre SC, Cady NC, Coates GW, Ober CK, Batt CA. "Bionanofabrication Polyhydroxyalkanoates (PHAs) Micro-/Nano-structures on Solid Surfaces and Its Applications in Nanobiotechnology." Oral Presentation at the 2nd ASM – IEEE EMBS Conference on Bio, Micro and Nanosystems, San Francisco, CA, January 15-18, 2006. Speaker for Section: Nano-Bio Interface at IEEE EMBS Conference on Bio, Micro and Nanosystems 2006. , Jan-2006
Articles in Peer-reviewed Journals	Niamsiri N, Bergkvist M, Delamarre SC, Cady NC, Coates GW, Ober CK, Batt CA. "Enhanced In Situ Bionanofabrication of Polyhydroxybutyrate (PHB) Micro-/Nano Structures on Solid Surfaces. " Biomacromolecules (Submitted on 4-13-06). , Aug-2006
Books/Book Chapters	Delamarre SC, Niamsiri N, Batt CA. "Bionanofabrication: a Tool for Creating Unique Structures Through In Vitro Polymer Synthesis." in "Microbial Bionanotechnology: Biological Self-Assembly Systems and Biopolymer-Based Nanostructures." Ed. B. H. A. Rehm. Norwich, U.K. : Horizon Scientific Press, 2006. Chapter 2., Mar-2006